

TCS349: Responsible and Explainable AI

Unit 2, Lecture 8

Data Protection in Practice: Minimisation, Anonymisation & Access Control

The concrete techniques that turn privacy principles into practice - deciding what data to keep, how to strip out identity, and who gets to see what remains.

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Minimisation



Anonymisation



Access Control



Worked Example

1. Core Data-Protection Principles

Lawfulness and fairness: personal data must be collected and used on a legitimate, disclosed basis.

Purpose limitation: data is used only for the purpose it was originally collected for, as covered in Lecture 16.

Data minimisation and storage limitation: only necessary data is collected, and it is kept only as long as needed.

Integrity and confidentiality: data must be protected against unauthorised access, loss or alteration.

2. Data Minimisation - Collect Only What's Needed

Data minimisation means collecting the smallest amount of personal data necessary to achieve a system's purpose.

Every additional data field collected is an additional point of privacy risk if the system is breached or misused.

Minimisation often improves a system's efficiency too, since less irrelevant data means less noise for a model to learn from.

A useful test: if a field were removed, would the system still achieve its stated purpose? If yes, it likely should not have been collected.

3. Anonymisation - Removing Identity from Data

Anonymisation is the process of altering data so that an individual can no longer be identified from it, directly or indirectly.

Properly anonymised data is generally considered outside the scope of the data-protection obligations that apply to personal data.

Anonymisation differs from pseudonymisation, which replaces identifiers with a code that can still be reversed with a separate key.

The strength of an anonymisation technique determines how much genuine privacy protection it actually provides.

4. Anonymisation Techniques - Masking & Generalisation

Data masking: replacing sensitive values with realistic but fake substitutes, such as a masked phone number.

Generalisation: reducing precision, such as replacing an exact age with an age range or a full address with just a city.

Both techniques keep the data usable for analysis while reducing how precisely a record can be tied to one person.

They are among the simplest anonymisation methods to apply, but also among the easiest to reverse if used alone.

5. Anonymisation Techniques - k-Anonymity & Differential Privacy

k-anonymity: ensuring each record is indistinguishable from at least $k-1$ other records on key identifying attributes.

Differential privacy: adding carefully calibrated statistical noise so individual records cannot be reliably distinguished in aggregate results.

Both are stronger, more formal guarantees than simple masking or generalisation, but require more careful implementation.

Differential privacy is increasingly used in large-scale statistics, including national census data releases.

6. The Re-Identification Risk

Anonymised data can sometimes be re-identified by combining it with other available datasets, even without any single direct identifier.

A well-known case showed that a specific individual's record could be re-identified from an anonymised dataset using just a few indirect attributes, such as ZIP code, birth date and gender.

This risk grows as more auxiliary data becomes publicly available over time, meaning anonymisation is not a permanent guarantee.

Because of this risk, anonymisation should be treated as one layer of protection, not a complete solution on its own.

7. Access Control - Limiting Who Can See Data

Access control restricts who inside or outside an organisation can view, use or export personal data.

The principle of least privilege means a person should only have access to the data strictly necessary for their role.

Access control matters even for well-anonymised data, since re-identification risk grows when data is combined without oversight.

It is the layer of protection that remains active for as long as the data itself is retained.

8. Access Control Mechanisms

Mechanism	What It Does
Role-Based Permissions	Grants access based on job role, not individual request
Audit-Logged Access	Records every access request for later review
Time-Limited Grants	Automatically expires access after a set period

9. Worked Example - Applying These Techniques Together

A health-research dataset, before and after applying minimisation, anonymisation and access control:

Field	Original Value	After Applying Techniques
Name	Ananya Sharma	Removed (minimisation)
Age	27	25-30 (generalised)
Pin Code	263136	263xxx (generalised)
Phone Number	98XXXXXXXX	Removed (minimisation)
Diagnosis	Type 2 Diabetes	Retained, access-controlled

10. Privacy-Aware Techniques Across the AI Lifecycle

Collection stage: data minimisation limits what enters the system in the first place.

Storage stage: anonymisation reduces the identifiability of data that must be kept.

Access stage: access control governs who can view or export data for as long as it exists.

Together, these three techniques form a layered defence rather than a single point of protection.